

rinku: Self-Verifiable URLs for Trustless Verification

Abstract. Today, most decentralized networks require external tooling to validate the most basic unit of their system. Sure, it's just an API call to a trusted node to validate a transaction as genuine, but what if it could be confirmed solely client-side, even *offline*? We propose a distributed network in which URLs serve as self-contained, cryptographic proofs needed for said verification. This would enable trustless confirmation without reliance on any external infrastructure or services. This URL carries not just transaction data, but its complete verification path - ancestry, signatures, and checkpoint anchors. Ultimately, *the link itself is the proof*. This paper focuses on the URL-native proof system that the rinku network leverages.

1. The Problem with Modern Verification

Traditional blockchain networks require some form of extraneous software for verification, such as a light client, node, an API, etc. Other lightweight solutions such as quorum-signed receipts also exist, but they require fetching a validator set from the chain. Same story for gossip witnessing, timestamp/anchoring, even zk proofs - all of these either rely on the live network, or external infrastructure. Further, additional latency is introduced due to this secondary confirmation process. One way or another, there's an issue with trust, overhead, delays, or overall reliability. An API may be inaccessible, network conditions unreliable, or hardware restrictions surface. In the end, these gaps create risk within the user experience, and overall assurances of the transactional process.

Ultimately, *some infrastructure* is required to be trusted & relied upon to provide proofs. However in rinku's case, verification becomes self-contained. Imagine point-of-sale that is immediately confirmable on the client itself, nearly instantaneous with absolute certainty?

2. URLs as Proofs

Instead of storing data on-chain and fetching proofs from nodes, we can encode proofs directly into URLs that are returned within a transaction's receipt:

For example

```
rinku://tx/{base64url(deflate(transaction + ancestry + checkpoint))}
```

A Rinku URL contains:

- The transaction data (sender, recipient, amount, signature)
- Ancestry chain back to a finalized checkpoint

- Checkpoint anchor (Merkle root, validator attestations)

3. How It Works

3.1 Transaction Encoding

Transactions are encoded as compressed JSON directly into the URL:

Transaction -> JSON -> DEFLATE -> Base64url -> URL

A single transaction URL is roughly 600 characters. Using 5 levels of ancestry (proving the transaction chains back to a checkpoint), URLs remain under 1,500 characters which fits inside a QR code.

3.2 Proof Structure

A proof bundle contains:

```
{
  "tx": {
    "from": "a1b2c3...",
    "to": "d4e5f6...",
    "amount": 1000000,
    "nonce": 42,
    "sig": "..."
  },
  "fromPubKey": "BASE64_ECDSA_P256_PUBLIC_KEY",
  "hash": "sha256(tx)",
  "parents": [/* recursive proof bundles */],
  "checkpoint": {
    "id": "cp_789",
    "merkleRoot": "...",
    "height": 1000,
    "blsAggregateSig": "...",
    "signerBitmap": "...",
    "validatorProof": { /* MerkleSumTree multi-proof */ }
  }
}
```

Key elements:

- fromPubKey - The sender's full ECDSA P-256 public key, enabling signature verification. The from field is its fingerprint (SHA-256 truncated to 40 hex chars)
- checkpoint - Contains the finality proof: BLS-aggregated signature from validators, a bitmap indicating which validators have signed, & a Merkle proof for the validator set

Each parent itself becomes a proof bundle, creating a recursive structure that traces back to a known checkpoint.

3.3 Verification Algorithm

```
function verify(proofUrl):
  bundle = decode(proofUrl)

  // 1. Verify pubkey matches fingerprint
  assert fingerprint(bundle.fromPubKey) == bundle.tx.from

  // 2. Verify transaction signature via the pubkey
  assert ecdsaVerify(bundle.fromPubKey, bundle.tx, bundle.tx.sig)

  // 3. Verify hash integrity
  assert sha256(bundle.tx) == bundle.hash

  // 4. Verify all parents
  for parent in bundle.parents:
    assert verify(parent)

  // 5. Verify checkpoint finality
  assert verifyCheckpointSignatures(bundle.checkpoint)

  return true
```

3.4 Data Availability

Important clarification: Self-provable URLs guarantee **verification** without infrastructure, but not **discovery**. The URL must reach the verifier through some sort of transport, such as:

- Shared URL via QR code, message, email etc
- URLs are stored in application databases
- Crawlers can follow parent references to reconstruct comprehensive history

3.5 The rinku Network

rinku is a DAG-based distributed network where:

- **Shared DAG structure** - Transactions reference 1-2 recent global **tip** transactions (the latest unconfirmed transactions across the network), weaving all network activity into a unified DAG
- **Nonce-based ordering** - Each account maintains the *classic* sequential nonce to prevent replay attacks & establish per-account ordering
- **Weight-based consensus** - Conflicts are resolved by cumulative weight (based on account age & stake), ensuring Sybil resistance

Finality via checkpoints:

Validators periodically create checkpoints that finalize batches of transactions. A checkpoint includes:

- A Merkle root committing to all finalized transactions
- A BLS-aggregated signature from $\geq 67\%$ of the validator set
- A commitment to the next validator set

Once a transaction is included in a checkpoint and confirmed, a self-provable URL is generated. The checkpoint's aggregated signature becomes the trust anchor embedded inside the proof.

4. Proof Profiles

Different use cases require different security/size tradeoffs:

Profile A: Receipt (~600 - 2,300 characters)

What it proves: Transaction is valid and chains to a prior checkpoint **Trust assumption:** Verifier trusts the checkpoint was correctly signed **Use case:** PoS receipts, payment confirmations

`rinku://tx/{payload}`

Profile B: Full Finality (~3,000 - 10,000 characters)

What it proves: Transaction is Merkle-included in a checkpoint signed by $\geq 67\%$ of validators **Trust assumption:** Verifier knows the validator set **Use case:** High value (or risk) settlements

`rinku://txp/{payload}`

Profile C: Self-Contained (~1,600 - 2,800 characters)

What it proves: Everything in Profile B, in addition to the validator set commitment itself **Trust assumption:** Trust anchor is minimal and can be pinned once (genesis or pinned checkpoint); subsequent proofs are offline-verifiable **Use case:** Fully offline verification, air-gapped systems, legal evidence

`rinku://sp/{payload}`

5. Size Analysis

Real-world measurements using high-entropy data (e.g signatures):

Proof Type	Transactions	URL Length
Single tx	1	~600 chars
2-depth ancestry	3	~940 chars
5-depth ancestry	6	~1,400 chars
10-depth ancestry	11	~2,200 chars

Platform Compatibility

Platform	Limit	Single tx	5-depth
QR Code (L)	2,953 bytes	x	x
QR Code (H)	1,273 bytes	x	
Browser URL	65KB+	x	x

note: QR capacity depends on QR encoding mode; base64url typically uses byte mode. For maximum density we recommend a QR-optimized encoding (e.g base45) for tx/sp receipts.

6. Trust Bootstrapping

A fresh verifier needs a trust anchor:

1. **Genesis trust** - Know the genesis validator set
2. **Checkpoint chain** - Each checkpoint commits to the next validator set
3. **Pinned checkpoint** - Trust a recent checkpoint from an out-of-band source

This mirrors TLS certificate chains - the proof is self-contained, but root trust must be established externally. Once bootstrapped, all subsequent proofs verify offline.

7. Why This Matters

7.1 Infrastructure Independence

Traditional flow (simplified):

User -> Trust Node -> Query Proof -> Verify with Node

Rinku flow:

User -> Receive URL -> Verify Locally

7.2 Portable Proofs

A Rinku URL can be:

- Printed as a QR code on a receipt
- Sent via SMS, email, or messaging app
- Embedded in a PDF or document
- Stored offline indefinitely

7.3 Offline-First

Verification works completely offline. This enables:

- PoS in areas with poor or no connectivity
- Air-gapped security systems
- Archival verification that withstands over large periods of time
- Cross-border payments without infrastructure

8. Cryptographic Primitives

The proof system uses standard, well-audited cryptography:

Component	Algorithm	Purpose
Transaction signatures	ECDSA P-256	Sender authorization
Hash function	SHA-256	Transaction identity, Merkle trees
Validator signatures	BLS12-381	Aggregated checkpoint attestation
Compression	DEFLATE	URL size reduction
Encoding	Base64url	URL-safe representation

ECDSA verification can use the Web Crypto API; BLS verification can use a WASM library (e.g blst)

9. Limitations

Size constraints: Complex proofs (30+ transactions) exceed QR capacity and require standard URLs

Bootstrap requirement: First-time verifiers need a trust anchor (genesis or pinned checkpoint)

URL mutability: URLs can be shared but not modified. Proof updates require re-generated URLs

Compression variability: Actual compression ratios depend on transaction content. High entropy

data compresses less

10. Conclusion

The rinku network demonstrates that transactional proofs can be fully self-contained within URLs. By encoding transaction data, ancestry chains, and checkpoint anchors directly into the URL, we can eliminate infrastructure dependencies for verification, and vastly improve the transactional experience for network participants.

The core innovation is architectural: treating URLs as the canonical proof object rather than as references to external state. This enables truly trustless, offline verification.

Appendix A: Encoding Specification

A.1 URL Format

`rinku://{profile}/{base64url(deflate(json))}`

Profiles:

- tx - Transaction receipt (Profile A)
- txp - Transaction with full ancestry (Profile A+)
- sp - Self-contained finality proof (Profile C)

A.2 Transaction Schema

```
interface Transaction {  
  from: string;    // 40-char hex fingerprint  
  to: string;      // 40-char hex fingerprint  
  amount: number;  // Smallest units (8 decimals)  
  fee: number;     // Gas fee  
  nonce: number;   // Sender sequence number  
  tipUrls: string[]; // 0-2 parent references  
  ts: number;      // Unix timestamp (ms)  
  sig: string;     // ECDSA P-256 signature  
}
```

A.3 Proof Bundle Schema

```
interface ProofBundle {  
  tx: Transaction;  
  fromPubKey: string; // Base64 ECDSA P-256 public key (SPKI format)  
  hash: string;       // SHA-256 hex of canonical tx JSON  
}
```

```

parents: ProofBundle[]; // Recursive ancestry
checkpoint: {
  id: string;
  merkleRoot: string;
  height: number;
  blsAggregateSig: string; // BLS12-381 aggregated signature
  signerBitmap: string; // Bitmap of which validators signed
  validatorProof: { // MerkleSumTree multi-proof
    signerLeaves: Array<{ pubKey: string; weight: number }>;
    auxiliaryNodes: string[];
    totalWeight: number;
  };
};
}

```

A.4 Canonical JSON

Transaction fields are serialized in deterministic order for consistent hashing: from, to, amount, fee, nonce, tipUrls, ts, sig. Canonical JSON = UTF-8, no whitespace, exact numeric encoding rules.

Appendix B: Reference Implementation

A minimal verifier in javascript:

```

async function verifyProofUrl(url) {
  // 1. Parse and decode url
  const [, profile, payload] = url.match(/rinku:\V\(\w+)\V(.+)/);
  const json = inflate(base64urlDecode(payload));
  const bundle = JSON.parse(json);

  // 2. Verify pubkey matches fingerprint
  const pubKeyBytes = base64Decode(bundle.fromPubKey);
  const fingerprint = (await sha256(pubKeyBytes)).slice(0, 40);
  if (fingerprint !== bundle.tx.from) {
    throw new Error('Public key does not match sender fingerprint');
  }

  // 3. Verify transaction signature using bundled pubkey
  const txBytes = canonicalize(bundle.tx);
  const pubKey = await crypto.subtle.importKey(
    'spki', pubKeyBytes,
    { name: 'ECDSA', namedCurve: 'P-256' },
    false, ['verify']
  );
  const valid = await crypto.subtle.verify(
    { name: 'ECDSA', hash: 'SHA-256' },
    pubKey,

```



```
base64Decode(bundle.tx.sig),
txBytes
);
if (!valid) throw new Error('Invalid signature');

// 4. Verify hash integrity
const hash = await sha256Hex(txBytes);
if (hash !== bundle.hash) throw new Error('Hash mismatch');

// 5. Verify parents
for (const parent of bundle.parents) {
  await verifyProofUrl(encodeBundle(parent));
}

// 6. Verify checkpoint finality (BLS aggregate signature)
return verifyCheckpointSignatures(bundle.checkpoint);
}
```

References

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 2. D. Boneh, B. Lynn, and H. Shacham. "Short Signatures from the Weil Pairing." Journal of Cryptology, 2004.
 3. R. Merkle. "A Digital Signature Based on a Conventional Encryption Function." CRYPTO 1987.
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